IDEATHON

MediScribe AI

- **Problem Statement Title:** Mitigating Clinician Burnout through Intelligent Documentation
- **Theme:** Health and BioTechnology (AI in Healthcare)
- **Team Name:** MediScribe AI

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PROBLEM IDENTIFIED

❖ Selected Theme / Problem

The Documentation Crisis: Clinicians spend nearly 50% of their time on EHR documentation, leading to 'Note Bloat' and severe burnout.

Detailed explanation of the proposed problem

- **Cognitive Overload:** Doctors juggle fragmented tools (notes, voice memos, history), breaking clinical focus.
- **Inefficiency:** Manual data entry from unstructured files (handwritten notes/PDFs) is slow and error-prone.
- **Lack of Synthesis:** Existing tools store data but cannot 'understand' it or synthesize diagnoses from multimodal inputs.

Literature Gap

- Current solutions are either rigid EHR systems or generic note-taking apps.
- Missing: A specialized 'Clinical OS' combining sketching, voice, and RAG-based analysis.

Need of Addressing the Problem (Desirability)

Automating 70% of documentation restores the doctor-patient relationship and reduces medical errors.

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IDEA TITLE

❖ Proposed Solution

MediScribe AI: A next-generation Clinical Journaling & Patient Management System powered by Multimodal AI.

Detailed explanation of the proposed solution

- **Unified Workspace:** 'Glassmorphic' interface combining Journaling, Patient Records, and AI tools.
- **Dr. Drishti (RAG Engine):** Instantly analyzes uploaded reports to answer clinical queries without hallucinations.
- **Dr. Doodle & Dhvani Sanchar:** Converts clinical sketches and voice dictations into structured digital records.
- **Searchy Gooper**: A persistent floating assistant for summarization and navigation.

How it addresses the problem

- **Centralizes** Voice, Text, and Vision into one dashboard.
- **Eliminates** context switching and enables grounded clinical retrieval.

Innovation and uniqueness

- **Zero-Hallucination**: RAG framework retrieves answers only from verified uploaded documents.
- **Multimodal Native**: Simultaneously processes audio, images (X-rays), and text.
- **Offline First**: Core features designed to function without constant internet dependence (Local DB).

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TECHNICAL APPROACH

Technologies to be used

- **Frontend:** React 18 (Vite), TypeScript, Tailwind CSS, Framer Motion.
- **Backend:** Node.js, Express.js, SQLite (MVP) / PostgreSQL (Production).
- **AI Core:** Google Gemini Pro (Reasoning), Gemini 1.5 Flash (Vision & Audio).
- **Security:** AES-256 Encryption for Patient Health Information (PHI).

Methodology and process for implementation

Input Layer

Clinician uploads PDF, speaks (Voice), or draws (Canvas).

↓

Processing Layer (Gemini AI + RAG)

Voice → Transcribed to Text.
PDF → Vectorized for Contextual Search.
Sketch → Analyzed for Medical Context.

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Output Layer

Structured Patient Record saved to Secure Database.

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FEASIBILITY AND VIABILITY

Analysis of feasibility

- **High Feasibility:** MVP is fully functional with real-time data persistence and AI integration.
- **Scalable Architecture:** Decoupled Frontend/Backend allows easy migration to Cloud (AWS/GCP).

Potential challenges and risks

- **Data Privacy:** Strict handling of sensitive Patient Health Information (PHI).
- **AI Latency:** Processing large medical PDFs in real-time.
- **Adoption:** Steep learning curve for non-tech-savvy doctors.

Strategies for overcoming challenges

- **Privacy:** HIPAA-eligible cloud providers (GCP) + Role-Based Access Control (RBAC).
- **Performance:** Intelligent caching and 'Gemini Flash' models for sub-second responses.
- **Adoption:** Apple-inspired intuitive UI to lower entry barriers.

THE LINK FOR MVP: https://ai.studio/apps/dbc6010e-037e-4e98-ac95-86ce86958116?fullscreenApplet=true

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IMPACT AND BENEFITS

Potential impact on target audience

- **Better Decisions:** AI-assisted retrieval ensures no critical data is missed.
- **Work-Life Balance**: Automating notes reduces burnout.
- **Professional Growth:** Integrated research tools help doctors stay updated.

Benefits of the solution

Social: Restores the doctor-patient relationship by removing screen distractions.

Economic: Optimizes clinic workflow, allowing for higher patient throughput.

Operational: Cuts data retrieval time by half; ensures secure, organized storage.

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MONETIZATION STRATEGY

Revenue sources

- **Freemium (Community Building):** Free basic access for medical students to build long-term loyalty.
- **SaaS Subscription (B2C):** 'Professional Tier' for individual doctors ($/month) offering unlimited AI queries.
- **Enterprise Licensing (B2B):** Custom contracts for Hospitals/Clinics including EHR integration.

Cost structure

- **API Usage:** Google Gemini Token consumption (Variable).
- **Infrastructure:** Cloud hosting (AWS/GCP) and Database storage.
- **Development:** Continuous security updates and AI model fine-tuning.

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COST ESTIMATION

Resources Required

- Manpower: 2 Full Stack Developers, 1 UI/UX Designer, 1 AI Engineer.
- Equipment: High-performance workstations, Tablets/iPads for testing 'Dr. Doodle'.

Tech Stack Budget (Initial Phase)

- Google Gemini API Credits: ~$500/month (Pilot Phase).
- Cloud Hosting (Vercel/Render): ~$50/month.
- Domain & SSL Security: ~$20/year.

Total MVP Budget: Low capital requirement (Software-focused development).

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RESEARCH AND REFERENCES

Details / Links of the reference

- Burnout Statistics: “Physician Burnout: The Root of the Problem” (National Academy of Medicine).
- RAG Architecture: “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks” (Lewis et al., 2020).
- Technical Docs: Google Gemini API Documentation & React.js Documentation.
- Design Philosophy: Apple Human Interface Guidelines (Design for Clarity & Deference).

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FRONTEND ARCHITECTURE: A CRAFTED EXPERIENCE

Core Framework: React 18 + Vite

- Selected for component-based reusability and optimized production builds.

Design System

- **Tailwind CSS**: Utility-first framework for rapid, consistent medical-grade styling.
- **Framer Motion**: Handles fluid transitions (modal scaling, slide-overs) to reduce cognitive load.
- **Lucide React**: Provides consistent, clean iconography.

Architecture

- **Single Page Application (SPA)**: Ensures seamless user interaction without page reloads.
- **Direct AI Integration**: Client-side handling of non-sensitive AI interactions for responsiveness.

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BACKEND & AI CORE: SECURE & SCALABLE

Backend API (Express.js)

- **Lightweight**: Handles CRUD operations for journals and patient history.
- **Modular**: Designed to swap data layers (Local -> Cloud) without rewriting logic.

Data Persistence

- **Current**: SQLite (Local file-based) for rapid MVP prototyping.
- **Roadmap**: Migration to PostgreSQL / Google Cloud Firestore for concurrent user scaling.

AI Integration Strategy

- **Decoupled**: Independent scaling of AI services from the core application.
- **Security**: Server-side validation of all API requests to protect tokens.

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TECHNICAL Q&A & FUTURE FUTURE ROADMAP

Q: How does the system scale with a file-based database?

A: The backend features a modular data access layer. Migration to PostgreSQL is a planned configuration change, not a rewrite.

Q: How do you prevent medical AI medical AI hallucinations?

A: Strict RAG Enforcement. The system prompt explicitly forbids external knowledge, forcing the AI to answer only from the uploaded PDF context.

Q: What is the strategy for HIPAA compliance? for HIPAA compliance?

A: Production will utilize HIPAA-eligible Google Cloud services, BAA signing, encryption at rest/transit, and strict Role-Based Access Control (RBAC).

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